# Hyperspectral Image Classification Using Geometric Spatial–Spectral Feature Integration: A Class Incremental Learning Approach

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***Abstract*— Hyperspectral image classification (HSIC) has attracted widespread attention due to its important application in environment alterations and geophysical disaster monitoring. However, surface cultivation is not static as time passes, which leads to different hyperspectral image (HSI) information collected from the same area at different time periods. Therefore, researchers are currently eager to construct an HSIC model that continuously acquires new classes of data. During the continuous learning process, the model is expected to not only be effective in extracting unique spatial–spectral features of the HSI but also ensure the ability to maintain the old classes' knowledge while learning new data. To achieve this purpose, we propose a method that is based on geometric spatial–spectral feature integration network with class incremental learning ($GS_2$FIN-CIL) framework in continuous learning to make the model adaptable to new classes' data and not overly forgetting the old classes' knowledge during the training process. We conduct extensive experiments with the proposed $GS_2$FIN-CIL method on widely used hyperspectral datasets, including Indian Pines (IP), PaviaU, and Salinas (SA). The experimental results show that our $GS_2$FIN-CIL method can achieve significantly improved results compared to current state-of-the-art class incremental learning (CIL) methods, allowing for efficient adaptation and utilization of spatial–spectral features in processing new classes of HSIs and alleviating the problem of catastrophic forgetting of learned old classes' knowledge. The $GS_2$FIN-CIL method could be successfully applied to the challenge of adding new classes' data in the HSIC task.**

***Index Terms*— Attention mechanism, class incremental learning (CIL), continuous learning, hyperspectral image classification (HSIC), spatial–spectral feature extraction.**

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## I. Introduction

HYPERSPECTRAL image classification (HSIC) is a pixel-level image classification task for ground objects, which is widely used in the field of remote sensing, including agricultural analysis, natural hazard detection, military surveillance, and mineral exploration [1], [2]. Compared to other optical images [3], [4], [5], the major characteristic of hyperspectral images (HSIs) lies on the coverage of substantial spatial–spectral features corresponding to surface crops.

In recent years, there have been numerous deep learning methods to extract the spatial–spectral features effectively [6], [7], [8], [9]. Considering the geometric characteristics of HSIs as 3-D data cube, most of the deep learning methods can only explore the spatial features while failing to explore the intrinsic connection between spatial and spectral features satisfactorily [10], [11], [12], [13]. To address this issue, many researchers have developed geometric deep learning for HSIC, to process hyperspectral data from a multidimensional perspective [14], [15], [16], [17]. Numerous studies have shown that the utilization of geometric deep learning on HSIC achieved promising results [18], [19], [20], [21], [22].

At present, the natural environment and human explorations have led to gradual changes in the land surface, which requires continuous learning of new classes of HSIs [23]. For example, during anthropogenic desertification improvement, the new classes of land surface need to be monitored to ensure the practical application of the improvement measures. Under the influence of natural disasters, the ground object classes will be subject to change. Timely detection is required for the new categories so that they would not impact the landscape aggressively and cause more serious geological hazards.

Facing the problem of adding new classes to HSIs, existing works have utilized class incremental learning (CIL) on HSIC task [23], [24], [25], [26]. CIL, as a research branch of continuous learning, enables to continue learning additional

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classes' data based on the old classes' knowledge. In contrast to traditional deep learning, there is no requirement to repeat the learned old classes' knowledge in this process. On the down side, the CIL model tends to be more biased toward the new classes of data added, thereby leading to an inevitable forgetting of what has been learned from the old classes. This is referred to as catastrophic forgetting. Summarizing, the following technical challenges remain for achieving efficient HSIC in continuous learning of new classes of HSIs.

1) How to extract useful features from HSIs and utilize them productively based on the incremental additional classes. It is well known that HSIs contain a multitude of spatial–spectral features at the pixel level. Accordingly, a CIL model is required to extract features with beneficial classification with only a small amount of old classes' data and limited amount of new classes' data, which has long been a challenge in the HSIC task.
2) How to alleviate the problem of catastrophic forgetting of the model's old classes' knowledge when continuously learning new classes from HSIs data. As much as possible, we aim to reduce the model's forgetting of learned old classes' knowledge and to decrease the model's proximate bias for new classes while retaining part of old classes' data.

To overcome the abovementioned challenges, we strive to develop a CIL method for HSIC, namely, the so-called $GS_2$FIN-CIL. To specify, the proposed $GS_2$FIN-CIL mainly adopts the geometric spatial–spectral feature integration network with class incremental learning ($GS_2$FIN-CIL) framework to achieve continuous learning for HSIC tasks. In particular, we design a geometric spatial–spectral feature integration attention module ($GS_2$FIAM) in the network. The proposed $GS_2$FIAM can effectively extract the geometric spatial–spectral features of HSIs and deeply combine the features at the 3-D geometric level to improve the classification results.

On top of that, we propose a balanced class knowledge (BCK) classifier in $GS_2$FIN-CIL and design a new loss function system called memory loss function system (MLFS) to address the catastrophic forgetting problem. To mitigate the proximate bias, the BCK classifier is adopted in the testing phase to correct the bias of the CIL model for the old classes and new classes' average features. The proposed MLFS applies angle loss and combines it with cross-entropy loss and distillation loss, in which the angle loss specifically acts on the previous model to alleviate the catastrophic forgetting problem when facing the new classes HSIs data.

The main contributions of this article are summarized as follows.

1) The $GS_2$FIN-CIL framework is proposed to be capable of implementing continuous learning for additional classes in the HSIC task. Specifically, we design the $GS_2$FIAM, which is customized for 3-D data cubes such as HSIs, to deeply extract and effectively joint spatial–spectral features at the space level, thereby improving the classification accuracy of HSIC.
2) In the $GS_2$FIN-CIL, the proposed BCK classifier enables balancing the weights of the current model and the previous model. It hence corrects the model's proximate bias when new classes of data arrive in the HSIC task, thus mitigating the problem that the classification results will be more biased toward the new classes.
3) The MLFS is designed in the CIL framework to alleviate the catastrophic forgetting problem. It mainly adopts three loss functions: cross-entropy loss, distillation loss, and angle loss. Among them, the angle loss focuses on the old model, which ensures that the model would not forget the old classes of knowledge excessively.

We perform extensive experiments to demonstrate that the proposed method can continuously learn the additional classes' information and achieves superior classification results compared to current state-of-the-art CIL methods. The final classification results on Indian Pines (IP), Pavia University (PU), and Salinas (SA) hyperspectral datasets are 91.14%, 90.06%, and 95.99%, respectively.

The remaining part of this article is organized as follows. Section II describes some recent research works on CIL and spatial–spectral feature attention mechanism for HSIC. Section III describes the proposed CIL framework $GS_2$FIN-CIL in detail. Section IV details the experimental setup and the experimental results on different hyperspectral datasets. Section V summarizes the analysis of the experimental results and concludes this article.

## II. Related Work

### A. Class Incremental Learning

On account of remarkably robust flexibility in learning new classes' knowledge and stability in remembering previous knowledge, CIL is widely utilized for HSIC task that requires processing of added classes [23], [27], [28]. Facing the challenge of learning new classes persistently, the outstanding performance of CIL has attracted the attention of many researchers [29]. Within the current research works on CIL, there are three main categories.

1) *Regularization-Based:* It mainly minimizes the impact of the model on the weights when learning new classes' information, e.g., EWC [30], PathInt [31], and MAS [32].
2) *Examplar-Based:* It stores a certain amount of previous task data as a way to prevent forgetting the old classes' knowledge that has been learned, e.g., iCarL [33] and End-to-End [34].
3) *Bias Correction:* It is specifically a problem that often occurs in CIL, referring to the bias of the model against the latest learned tasks, e.g., BiC [35], LUCIR [36], and WA [37].

In recent years, CIL has been increasingly introduced to HSIC task to continuously learn new classes' information [38], [39], [40]. Bai et al. [38] proposed a linear programming CIL classifier (LPILC) to implement the challenge of learning new classes of HSIs. Xue et al. [39] designed an incremental dictionary learning (IDL) method on HSIC that enables CIL by iteratively introducing incremental samples into the dictionary and feeding the last round of incremental dictionary to the model to produce the classification map. The methods

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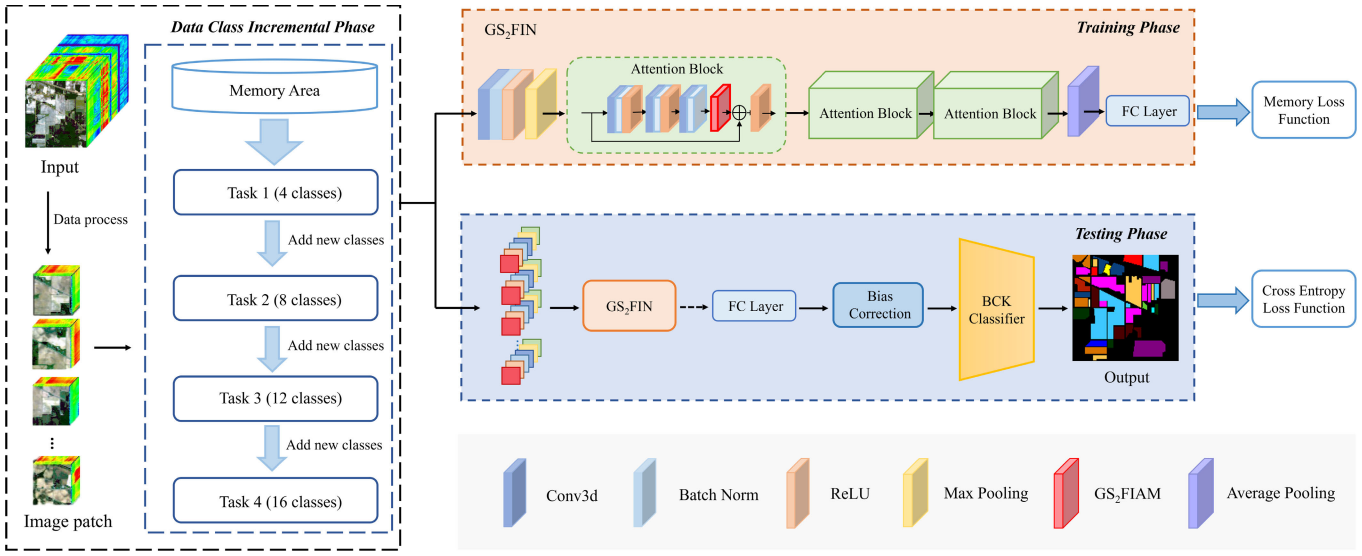



Fig. 1. Proposed CIL framework $GS_2FIN$-CIL, including the data class incremental phase with the example of IP dataset.

mentioned above have effectively demonstrated that CIL can achieve the goal of continuously learning new information from HSIs and obtain excellent classification results.

### B. Geometric Spatial–Spectral Feature Attention Mechanism

Geometric deep learning [41] draws widespread attention because of its ability to maximize the utilization of the collected information from the perspective of the data itself. It is crucially important for the network to be able to profoundly extract and utilize the features in HSIs when continuously learning data. Accordingly, the attention mechanism has been developed rapidly and has been widely adopted in recent decades due to their powerful feature extraction capabilities, such as squeeze-and-excitation (SE) [42], bottleneck attention module (BAM) [43], convolutional block attention module (CBAM) [44], and coordinate attention [45]. Many studies have shown that these attention mechanism modules have achieved good performance in extracting the spatial–spectral features of HSIs [46], [47], [48], [49], [50], [51]. Xu et al. [46] proposed a novel residual spatial attention kernel generation network (RSAKGN) to extract distinguished semantic features on HSIs. Zhu et al. [47] proposed a residual spectral–spatial attention network (RSSAN), which can suppress the useless information of the central pixel and adaptively enhance the useful spectral information that makes a significant contribution to the classification. Zhang et al. [51] proposed a spectral–spatial attention network (SSAN) that combined the 3-D CNN with spatial attention mechanism to capture discriminative spectral–spatial features from the HSI cubes. Liu et al. [50] specially tailored scaled dot-product central attention (SDPCA) for HSI and proposed a central attention network (CAN) to focus on extracting neighbor information that is similar to the current query pixel.

## III. Methodology

In this article, the $GS_2FIN$-CIL method is proposed, which primarily adopts the CIL framework to address the challenge of adding new classes that appears in the HSIC task when facing factors such as environmental changes. Specifically, the method is separated into two phases: the training phase and the testing phase. The overall framework of the $GS_2FIN$-CIL method is shown in Fig. 1.

1) *Training Phase:* In the training phase, the $GS_2FIN$ is designed to extract the integrated spatial–spectral features of HSIs from the 3-D level comprehensively. To alleviate the problem of excessive forgetting of the model, the new MLFS is proposed, which could enhance the memory level of the model for the previous knowledge.
2) *Testing Phase:* In the testing phase, we use the bias correction algorithm and introduce the designed BCK classifier to solve the proximate bias, which alleviates the catastrophic forgetting problem that the $GS_2FIN$-CIL method encounters when continuously learning new knowledge.

We describe the proposed method on hyperspectral datasets in Section III-A. In Section III-B, we focus on the designed $GS_2FIN$ framework and describe the designed $GS_2FIAM$ attention mechanism module. Specifically, we explain the proposed BCK classifier in Section III-C. Also, in Section III-D, the designed MLFS is presented in detail.

### A. Class Incremental Data Updation on HSIC

To continuously learn new classes' information for HSIs, CIL can be well utilization, which is the main framework of the proposed $GS_2FIN$-CIL method. Taking the hyperspectral dataset of IP dataset for example, which has 16 classes of ground truth, the total task of distinguishing these 16 classes is divided into four classification tasks, each with four additional classes in turn: task 1 (four classes), task 2 (eight classes), task 3 (12 classes), and task 4 (16 classes). To specify, the process in the incremental stage of data classes is shown in Fig. 2, where $M$ is the frozen memory area of data storage, which is the same as the memory area in Fig. 1.

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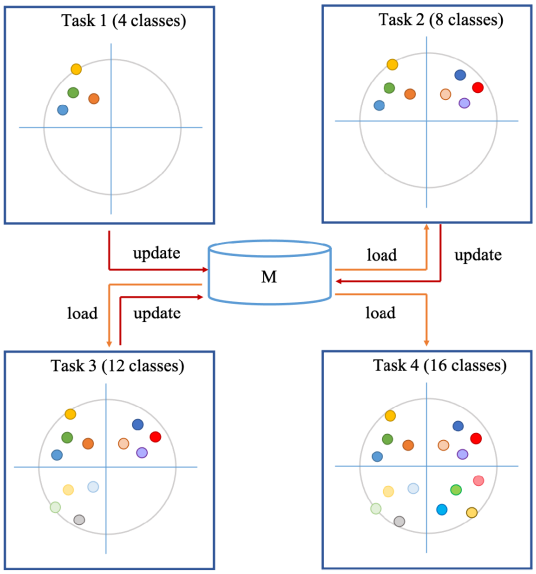



Fig. 2. Detailed process of the data class incremental phase with the example of IP dataset.

**Algorithm 1** Data Updation in $GS_2FIN$-CIL

**Input:** $X = (X^s, \ldots, X^t)$ // new classes exemplar data
**Input:** $K$, $\theta_{old}$ // memory size, old model
**require** $P^{s-1} = (P_1, \ldots, P_{s-1})$ // old exemplar sets
**require** $\theta_{new}$ // new model trained on t classes
$m \leftarrow K/t$ // number of exemplars selected per class
$\theta_{new} \leftarrow$UpdateCurrentModule$(X, P^{s-1}, m, \theta_{old})$
**for** $i = 1, \ldots, s-1$ **do**
$P_i \leftarrow$SaveExemplarSets$(P_i, m)$
**end for**
**for** $i = s, \ldots, t$ **do**
$P_i \leftarrow$UpdateExemplarSets$(X, m, \theta_{old})$
**end for**
**Output:** $P \leftarrow (P_1, \ldots, P_t)$ // new exemplar sets

The proposed method updates the data of the current task to the memory area when the current classification task is completed. To ensure that the model's memory for the previous knowledge would not be catastrophically forgotten, a portion of the data from the latest task is loaded when proceeding to the next class incremental task. Simultaneously, adding quantity of new classes' data $X = (X^s, \ldots, X^t)$, the combination of these data and old classes' data is taken as the training data for the current class incremental task. After this class incremental task is completed, the new exemplar set $P$ is updated to the memory area again as the training set of the old data for the next class incremental task. The data updation algorithm in the proposed method is shown in Algorithm 1.

### *B. Geometric Spatial–Spectral Feature Integration Network*

Aiming to continuously acquire information about additional classes by extracting superior useful spatial–spectral features in HSIs, the feature extraction network so-called $GS_2FIN$ is designed. The specific composition of the $GS_2FIN$ and attention blocks is shown in Fig. 1. The image patch $X$ inputs into the $GS_2FIN$, first sequentially through the 3-D convolution layer, batch normalization layer, Relu layer, and max-pooling layer for initial feature extraction, followed by three attention blocks for deep spatial–spectral feature integration extraction, and finally outputs to the fully connected layer through the average-pooling layer.

Specifically, in the attention block, the core component is the $GS_2FIAM$ module is designed for the 3-D cube of HSIs. The specific structure of $GS_2FIAM$ is shown in Fig. 3.

The input is a 3-D tensor patch $\mathcal{X} \in \mathbb{R}^{H\times W\times C}$ in HSIC, where $H$ and $W$ are the height and width in the spatial dimension, respectively, and $C$ is the number of spectral bands. The patch is able to obtain regarding the distinct dimensional patch $\mathcal{X}_c \in \mathbb{R}^{1\times 1\times C}$, $\mathcal{X}_w \in \mathbb{R}^{1\times W\times 1}$, and $\mathcal{X}_h \in \mathbb{R}^{H\times 1\times 1}$ on the geometric level of the HSI patch after averaging pooling layers from $XY$, $YZ$, and $XZ$ plane perspectives.

Since the information between spatial-level pixels correlates quite intensely, we performed the following operations in order to extract the effective spatial features $\mathcal{F}_{wh}$:

$$\mathcal{X}_{wh} = \text{Concat}\left(\mathcal{X}_w, \mathcal{X}_h^T\right) \tag{1}$$

$$\hat{\mathcal{F}}_{wh} = \delta\left(\text{BN}\left(\mathcal{X}_{wh}^*\right)\right) \tag{2}$$

where $\hat{\mathcal{F}}_{wh}$ denotes the preliminary extracted features, $\delta$ refers to ReLU function, and BN$(\cdot)$ and $*$ refer to batch normalization and convolution operation, respectively. Namely, $\mathcal{X}_{wh}^*$ denotes $\mathcal{X}_{wh}$ after convolution operation

$$\mathcal{F}_w = \sigma\left(\left(\text{Split}\left(\hat{\mathcal{F}}_{wh}\right)\right)^*\right) \tag{3}$$

$$\mathcal{F}_h = \sigma\left(\left(\hat{\mathcal{F}}_{wh}\right)^*\right) \tag{4}$$

$$\mathcal{F}_{wh} = \mathcal{X} \otimes \mathcal{F}_w \otimes \mathcal{F}_h \tag{5}$$

where $\sigma$ denotes the sigmoid function and $\otimes$ refers to the tensor product calculation. With the abovementioned steps, we eventually obtained the better profound spatial features $\mathcal{F}_{wh}$.

To explore the latent correlations between the spectral dimension and the spatial dimension, resemblance measures are employed to obtain the integration features of the spectra with different spatial angles $\mathcal{F}_{cw}$ and $\mathcal{F}_{ch}$

$$\mathcal{F}_{cw} = \mathcal{X} \otimes \mathcal{F}_c \otimes \mathcal{F}_w \tag{6}$$

$$\mathcal{F}_{ch} = \mathcal{X} \otimes \mathcal{F}_c \otimes \mathcal{F}_h \tag{7}$$

where $\mathcal{F}_c = \mathcal{X}_c$.

Summarizing, the features at various scales are deeply integrated to extract our ideal geometric spatial–spectral output feature $\mathcal{F}_{\text{out}}$

$$\mathcal{F}_{\text{out}} = \text{BN}\left(\left(\text{Concat}\left(\mathcal{F}_{wh}, \mathcal{F}_{cw}, \mathcal{F}_{ch}\right)\right)^*\right). \tag{8}$$

Notably, the convolution operation uses convolution kernel and stride, which are both 1 in $GS_2FIAM$.

The $GS_2FIN$ is capable of adaptively learning the spatial–spectral features of HSIs from a 3-D geometric perspective in the face of limited additional new classes' data. This facilitates the model to maintain a superior learning ability both for previous knowledge and new knowledge, which achieves the purpose of efficient continuous learning.

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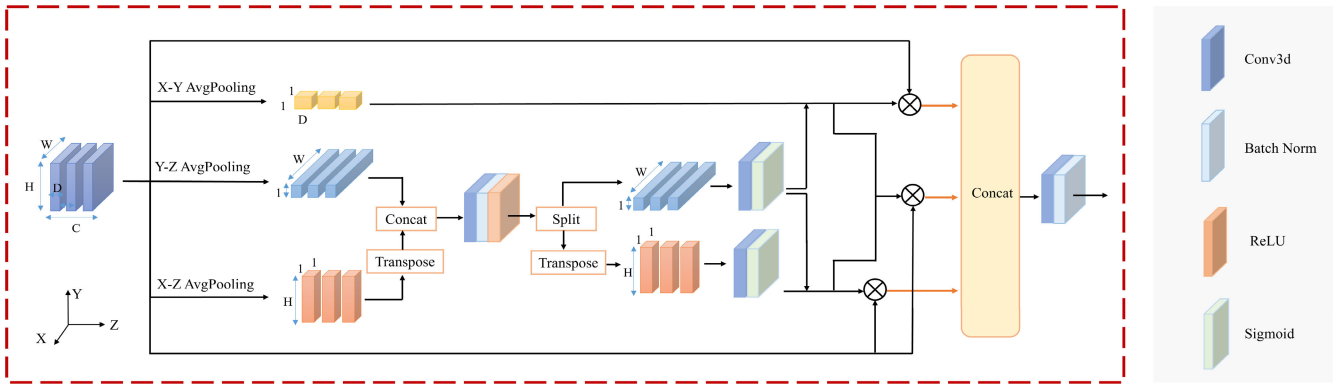



Fig. 3. Structure of attention module GS$_2$FIAM.

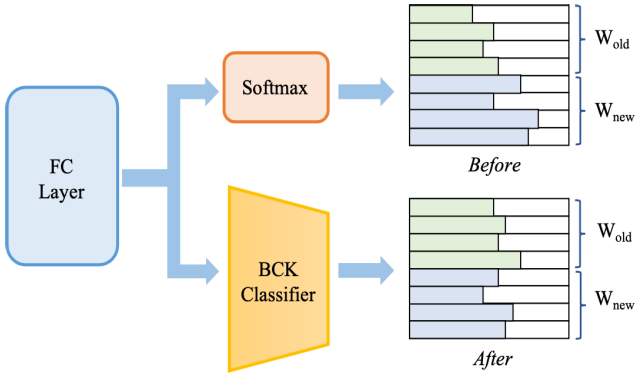



Fig. 4. Illustration of the weighs for old and new classes of the BCK classifier in CIL.

### C. BCK Classifier

In the testing phase, to focus on alleviating the problem of proximate bias that occurs in CIL, the BCK classifier is designed. The model inevitably pays higher attention to the new classes when encountering new classes' data, which results in the learned new classes weights $W_{\text{new}}$ unequally exceeding the old classes weights $W_{\text{old}}$. This situation impacts the accuracy of the model on the current classification task.

Uniquely, the BCK classifier is able to balance the learnable classes' knowledge of the model, which alleviates the proximate bias of the model for new classes as mentioned above. In contrast to the traditional softmax classifier, the utilization of the BCK classifier for the test set is to ensure that our model still performs effectively in generalizing to completely unseen data. The before-and-after effect of using the BCK classifier is shown in Fig. 4.

The bias of the model for various classes is explored to employ more reasonable measures in the BCK classifier to balance the learnable previous knowledge and new classes' knowledge. In the fully connected layer, each row of the input represents a category. Typically, the output vector $y_i$ of the input $i$th category is

$$\begin{aligned} y_i &= W_i x + b_i \\ &= \|W_i\| \cdot \|x\| \cdot \cos(W_i, x) + b_i \end{aligned} \tag{9}$$

where $x$ is the input vector, $W_i$ is the feature vector of the $i$th category, which can be considered as the learnable parameters, and $b_i$ is the constant bias. $\|\cdot\|$ represents the standardization of the vector. $\cos(W_i, x)$ is the cosine angle between $W_i$ and $x$, which is the offset of the model for the current input vector $x$ and the label vector.

In the case that the model would be biased toward the new classes, the cosine angle bias should be focused in (9). $\|x\|$ is the input vector itself, which has little impact on the specific prediction of the model classification. The cosine similarity between the different classes is

$$s_{ij} = \cos\left(W_i, W_j\right) = \frac{W_i W_j^T}{\|W_i\| \left\|W_j\right\|}. \tag{10}$$

The prediction results of the model for each class depend significantly on its prediction bias. With (10), we could calculate the balance values that the model needs for the old classes weights $\triangle W_{\text{old}}$ and new classes weights $\triangle W_{\text{new}}$, respectively

$$\triangle W_{\text{old}} = \text{Mean}(\cos(W_{\text{old}}, W_{\text{old}}))/(t - t_{\text{new}}) \tag{11}$$

$$\triangle W_{\text{new}} = \text{Mean}(\cos(W_{\text{new}}, W_{\text{new}}))/t_{\text{new}} \tag{12}$$

where Mean$(\cdot)$ represents the mean calculation, $t$ refers to the number of classes in the current classification task, and $t_{\text{new}}$ refers to the number of classes added in the new task.

Ultimately, the old classes weights $\hat{W}_{\text{old}}$ and the new classes weights $\hat{W}_{\text{new}}$ after balancing class knowledge are given in the following equations:

$$\hat{W}_{\text{old}} = W_{\text{old}} + \triangle W_{\text{old}} \tag{13}$$

$$\hat{W}_{\text{new}} = W_{\text{new}} - \triangle W_{\text{new}}. \tag{14}$$

Accordingly, the final output features of the BCK classifier are

$$o = \sigma\left(\text{Concat}\left(\hat{W}_{\text{old}}, \hat{W}_{\text{new}}\right)\right) \tag{15}$$

where $\sigma$ denotes the sigmoid function.

The BCK classifier regards the bias of the model for each class of old and new classes as balanced values, adjusting the learnable knowledge to alleviate the proximate bias.

### D. Memory Loss Function System

To address the problem of catastrophic forgetting effectively, we construct a new loss function system MLFS for the training phase of CIL. In the process of continuously learning new knowledge, the model loses some of the acquired knowledge, which is very similar to the human learning experience.

MLFS keeps the perspective on iterative updating of weights and aims at adaptively preserving the class bias of the previous

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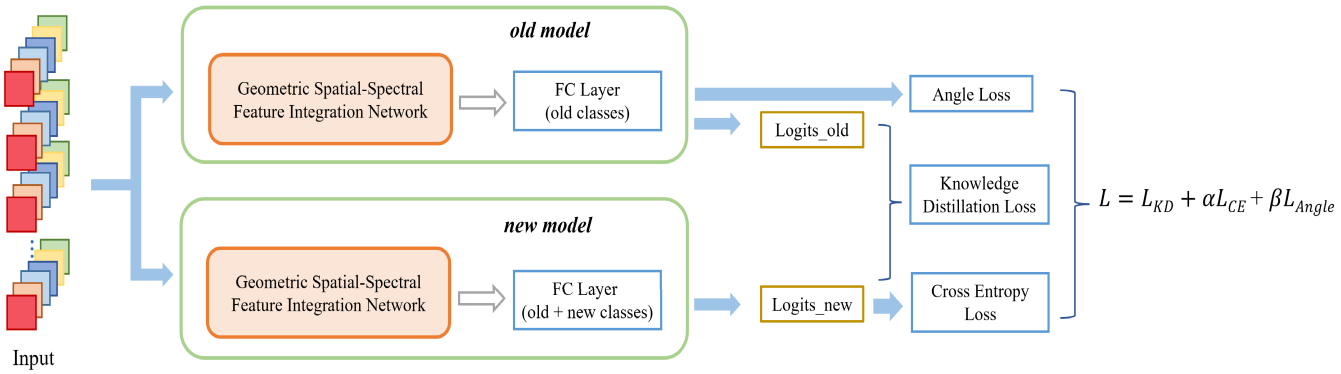



Fig. 5. Structure of the proposed new loss function system MLFS.

model in the process of continuous learning, which is so-called old knowledge memory. It partly avoids the forgetting of the learned knowledge by the new model. The loss functions of two stages in the GS$_2$FIN-CIL method are shown in Algorithm 2.

In the training phase, MLFS skillfully applies three loss functions and operates on the previous and current models simultaneously. The proposed MLFS structure is shown in Fig. 5. The knowledge distillation loss adaptively compares the $\text{Logits}_{\text{old}}$ with $\text{Logits}_{\text{new}}$ from the fully connected layer output. The cross-entropy loss is calculated using the new classes of logits. The cross-entropy loss and knowledge distillation loss are given in the following equations:

$$L_{\text{CE}} = -\sum_{i=s+1}^{t} y[i] \cdot \log\left(\text{Logits}_{\text{new}}[i]\right) \tag{16}$$

$$L_{\text{KD}} = D_{\text{KL}}\left(\text{Logits}_{\text{new}} \parallel \text{Logits}_{\text{old}}\right) \tag{17}$$

where, for $(x, y) \in P$, $\text{Logits}_{\text{old}} = \sigma(\Theta_{\text{old}}(x))$ and $\text{Logits}_{\text{new}} = \sigma(\Theta_{\text{new}}(x))$ are the output classification weights of the data on the previous model and the current model, respectively.

Compared to the traditional cross-entropy loss, the angle loss requires no correspondence between the input vector and the label vector. Instead, it is sufficient to directly operate on the previous model and obtain the old classes' bias of the model in order to memorize the previous knowledge. The ideology of angle loss to obtain class bias is given in (10). The final angle loss is calculated as

$$L_{\text{Angle}} = \text{Mean}(\cos(\Theta_{\text{old}})). \tag{18}$$

Finally, the overall loss function $L_{\text{MLFS}}$ can be summarized as

$$L_{\text{MLFS}} = L_{\text{KD}} + \alpha L_{\text{CE}} + \beta L_{\text{Angle}} \tag{19}$$

where $\alpha = t_{\text{new}}/t$, $\beta = 1 - \alpha$, $t_{\text{new}}$ denotes the number of classes added in the new task, and $t$ denotes the number of classes in the current classification task.

In the testing phase, the bias correction algorithm is used in front of the BCK classifier, which is a linear layer that can simply bias correct the output of the fully connected layer. The formula is given in the following equation:

$$\text{BiasCorrection}(x_k) = \begin{cases} x_k, & 1 \le k \le s-1 \\ \lambda x_k + \mu, & s \le k \le t \end{cases} \tag{20}$$

**Algorithm 2** MLFS

**Input:** $X = (X^s, \ldots, X^t)$, $P^{s-1} = (P_1, \ldots, P_{s-1})$ // new classes exemplar data, old exemplar sets
**Input:** $K$, $\theta_{old}$ // memory size, old model
**require** $\theta_{new}$ // new model trained on t classes
$m \leftarrow K/t$ // number of selected exemplars per class
$\theta_{new} \leftarrow$UpdateCurrentModule($X$, $P^{s-1}$, $m$, $\theta_{old}$)
**for** $(x, y) \in P$ **do**
**Training Phase:**
$o_{old} = \Theta_{old}(x)$
$o_{new} = \Theta_{new}(x)$
$Logits_{old} = \sigma(o_{old}) = softmax(o_{old})$
$Logits_{new} = \sigma(o_{new}) = softmax(o_{new})$
$p_{old} = Parameter(\Theta_{old})$
calculate cross entropy loss $L_{CE}$ (Eq. 16)
calculate distillation loss $L_{KD}$ (Eq. 17)
calculate angle loss $L_{Angle}$ (Eq. 18)
**Return** $L_{Train} = L_{MLFS} = L_{KD} + \alpha L_{CE} + \beta L_{Angle}$
**Testing Phase:**
$x \leftarrow BiasCorrection(x_k)$ (Eq. 20)
$o' = \Theta_{new}(x)$
$o'_{out} = \text{BCK}(o')$
calculate cross entropy loss $L_{CE}$ (Eq. 16)
**Return** $L_{Test} = L_{CE}$
**end for**
**Output:** $L_{Train}$, $L_{Test}$

where $x_k$ denotes the output logits of the $k$th class, $\lambda$ and $\mu$ denote the bias parameters on the new class, and $(s, t)$ denotes the number range of the new classes.

## IV. Experiments

### A. Hyperspectral Datasets

In this article, three classical hyperspectral datasets are mainly used for the experiments: IP, PU, and SA. The details of these datasets are described as follows.

1) *IP:* This dataset primarily consists of 145 × 145 pixels and 224 spectral bands. The scenes were collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor at the IP test site in Northwest Indiana, with two-thirds of the composition being agriculture and the rest being forest or natural vegetation. The

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TABLE I
SPECIFIC NUMBER OF EACH CLASS IN THE TRAIN SET AND TEST SET OF THE IP DATASET

| Class | Name | Train | Test |
|---|---|---|---|
| 1 | Alfalfa | 4 | 42 |
| 2 | Corn-notill | 142 | 1286 |
| 3 | Corn-mintill | 84 | 748 |
| 4 | Corn | 23 | 214 |
| 5 | Grass-pasture | 48 | 435 |
| 6 | Grass-trees | 73 | 657 |
| 7 | Grass-pasture-mowed | 2 | 26 |
| 8 | Hay-windrowed | 47 | 431 |
| 9 | Oats | 2 | 18 |
| 10 | Soybean-notill | 97 | 875 |
| 11 | Soybean-mintill | 245 | 2210 |
| 12 | Soybean-clean | 59 | 534 |
| 13 | Wheat | 20 | 185 |
| 14 | Woods | 126 | 1139 |
| 15 | Buildings-Grass-Trees-Drives | 38 | 348 |
| 16 | Stone-Steel-Towers | 9 | 84 |
| | Total | 1019 | 9232 |

ground truth contains a total of 16 classes, and both are independent.

2) *PU:* This dataset primarily consists of 610 × 610 pixels and 103 spectral bands. The scenes were collected by the Reflective Optics Spectrographic Imaging System (ROSIS) sensor over Pavia, Northern Italy, and mainly covered the spectral information of the University of Pavia. The ground truth contains a total of nine classes.
3) *SA:* This dataset primarily consists of 610 × 340 pixels and 224 spectral bands. Scenes are collected by AVIRIS sensors over the SA Valley, California, and include bare soil, garden fields, and so on. The ground truth contains a total of 16 classes.

TABLE II
SPECIFIC NUMBER OF EACH CLASS IN THE TRAIN SET AND TEST SET OF THE PU DATASET

| Class | Name | Train | Test |
|---|---|---|---|
| 1 | Asphalt | 199 | 6434 |
| 2 | Meadows | 559 | 18090 |
| 3 | Gravel | 62 | 2037 |
| 4 | Trees | 91 | 2973 |
| 5 | Painted metal sheets | 40 | 1305 |
| 6 | Bare Soil | 150 | 4879 |
| 7 | Bitumen | 39 | 1291 |
| 8 | Self-Blocking Bricks | 110 | 3572 |
| 9 | Shadows | 28 | 919 |
| | Total | 1278 | 41500 |

TABLE III
SPECIFIC NUMBER OF EACH CLASS IN THE TRAIN SET AND TEST SET OF THE SA DATASET

| Class | Name | Train | Test |
|---|---|---|---|
| 1 | Brocoli-green-weeds-1 | 40 | 1969 |
| 2 | Brocoli-green-weeds-2 | 74 | 3652 |
| 3 | Fallow | 39 | 1937 |
| 4 | Fallow-rough-plow | 27 | 1367 |
| 5 | Fallow-smooth | 53 | 2625 |
| 6 | Stubble | 80 | 3881 |
| 7 | Celery | 71 | 3508 |
| 8 | Grapes-untrained | 225 | 11046 |
| 9 | Soil-vinyard-develop | 124 | 6079 |
| 10 | Corn-senesced-green-weeds | 65 | 3213 |
| 11 | Lettuce-romaine-4wk | 21 | 1047 |
| 12 | Lettuce-romaine-5wk | 38 | 1889 |
| 13 | Lettuce-romaine-6wk | 18 | 898 |
| 14 | Lettuce-romaine-7wk | 21 | 1049 |
| 15 | Vinyard-untrained | 145 | 7123 |
| 16 | Vinyard-vertical-trellis | 36 | 1771 |
| | Total | 1077 | 53054 |

### B. Experimental Settings

*1) Data Preprocessing:* Based on the different data sizes of these three datasets in the experiment, we selected different ratios in dividing the training and test sets of each dataset, with 10% of the training set in IP dataset, 3% of the training set in PU dataset, and 2% of the training set in SA dataset. To ensure the reliability of the experimental results, the samples were selected randomly from each class. The specific numbers of training and test sets after the three datasets were divided according to the proportions shown in Tables I–III.

Before the data enter the network, we use the PCA operation to reduce the spectral dimension to 5 and divide the data block into patches in the form of central pixels, whose patch size is 48. Summarizing, the size of the processed 3-D data block is: 5 × 48 × 48.

*2) Parameter Settings:* The memory size is set to 480, which is the storage capacity of the old classes' data and the new classes' data. In the training phase, the optimizer is stochastic gradient descent (SGD), the learning rate is 0.01, the momentum is 0.9, and the weight decay is $2e^{-4}$. In the test phase for the bias correction operation, the selected optimizer is Adam, and the learning rate is 0.1. In the network, the 3-D convolutional layer and the max-pooling layer are used with a convolutional kernel of 3, a stride size of 1, and a padding of 1.

*3) Operation Environment:* The experiments are running on Linux system. GPU is Nvidia 1070 Ti, and the running environment version is summarized as follows: python 3.8, pytorch 1.12.0, torchvision 0.13.0, and numpy 1.23.4.

### C. Experimental Results

In this article, the $GS_2$FIN-CIL method performs abundant experiments on three classical hyperspectral datasets. Considering the reality that the sample space is not infinite in the continuous process of learning new data, the overall storage capacity of the samples is 480. In order to ensure the reliability of the experimental results, the experimental results obtained are the average values after ten experiments. The proposed method's classification results for class incremental tasks on the three datasets are shown in Tables IV–VI.

The classification result graphs on class incremental tasks will be shown in the comparison experiments in Section IV-D. In the tables of classification results, the value in parentheses indicates the value in the current task that is lower relative to

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TABLE IV
CLASSIFICATION RESULTS OF $GS_2$FIN-CIL METHOD ON THE IP DATASET

| Class | task 1 (4 classes) | task 2 (8 classes) | task 3 (12 classes) | task 4 (16 classes) |
|---|---|---|---|---|
| Alfalfa | 98.07±2.57 | 91.62±13.46 | 82.47±27.36 | 91.75±7.32 |
| Corn-notill | 97.34±3.845 | 96.62±0.78 | 92.22±2.26 | 90.73±1.75 |
| Corn-mintill | 98.63±1.25 | 92.31±3.53 | 92.62±2.87 | 91.18±2.89 |
| Corn | 96.36±1.46 | 94.14±3.80 | 95.56±2.53 | 95.25±2.96 |
| Grass-pasture | | 89.90±3.44 | 89.25±3.59 | 93.07±2.82 |
| Grass-trees | | 93.66±2.40 | 90.45±3.52 | 89.35±3.29 |
| Grass-pasture-mowed | | 63.78±12.67 | 89.47±29.87 | 90.67±11.86 |
| Hay-windrowed | | 98.26±0.81 | 99.33±0.63 | 95.32±1.80 |
| Oats | | | 47.25±14.81 | 95.00±10.67 |
| Soybean-notill | | | 90.34±2.16 | 90.28±1.83 |
| Soybean-mintill | | | 94.53±1.06 | 94.26±1.25 |
| Soybean-clean | | | 92.97±3.18 | 89.71±3.63 |
| Wheat | | | | 64.89±4.55 |
| Woods | | | | 94.56±1.89 |
| Buildings-Grass-Trees-Drives | | | | 90.57±3.41 |
| Stone-Steel-Towers | | | | 63.59±4.96 |
| OA | 97.56±2.62 | 94.07±1.01 (3.49) | 92.57±1.47 (1.50) | 91.14±0.59 (1.43) |
| AA | 97.60±1.17 | 90.04±2.67 (7.56) | 88.04±4.85 (2.00) | 88.72±1.16 (0.68↑) |
| Kappa | 95.63±4.86 | 92.53±1.26 (3.10) | 91.13±1.74 (1.40) | 89.91±0.66 (1.22) |

TABLE V
CLASSIFICATION RESULTS OF $GS_2$FIN-CIL METHOD ON THE PU DATASET

| Class | task 1 (3 classes) | task 2 (6 classes) | task 3 (9 classes) |
|---|---|---|---|
| Asphalt | 97.56±1.75 | 80.71±4.18 | 80.09±2.20 |
| Meadows | 99.69±0.23 | 97.01±2.06 | 97.68±0.39 |
| Gravel | 95.47±3.13 | 87.57±8.74 | 86.95±4.68 |
| Trees | | 62.87±9.34 | 67.78±4.96 |
| Painted metal sheets | | 98.03±2.38 | 86.35±0.41 |
| Bare Soil | | 96.35±2.46 | 98.49±0.83 |
| Bitumen | | | 88.17±2.65 |
| Self-Blocking Bricks | | | 90.04±1.27 |
| Shadows | | | 49.61±5.14 |
| OA | 98.82±0.51 | 90.04±1.96 (8.78) | 90.06±0.52 (0.02↑) |
| AA | 97.57±1.32 | 87.09±1.87 (10.48) | 82.80±0.87 (4.29) |
| Kappa | 97.51±1.05 | 85.41±2.86 (12.10) | 86.82±0.68 (1.41↑) |

TABLE VI
CLASSIFICATION RESULTS OF $GS_2$FIN-CIL METHOD ON THE SA DATASET

| Class | task 1 (4 classes) | task 2 (8 classes) | task 3 (12 classes) | task 4 (16 classes) |
|---|---|---|---|---|
| Brocoli-green-weeds-1 | 99.87±0.24 | 98.57±0.90 | 98.35±1.42 | 96.55±1.25 |
| Brocoli-green-weeds-2 | 99.85±0.26 | 98.83±0.71 | 97.12±0.88 | 99.46±0.46 |
| Fallow | 95.51±2.14 | 95.43±2.27 | 96.68±1.05 | 96.46±2.11 |
| Fallow-rough-plow | 99.46±0.83 | 97.53±1.77 | 95.65±1.98 | 95.08±2.36 |
| Fallow-smooth | | 95.53±2.09 | 94.75±2.21 | 94.53±1.53 |
| Stubble | | 98.53±0.52 | 95.54±1.62 | 95.78±1.07 |
| Celery | | 97.79±0.98 | 98.41±1.23 | 98.24±0.64 |
| Grapes-untrained | | 99.09±1.38 | 99.58±0.28 | 98.89±0.23 |
| Soil-vinyard-develop | | | 99.62±0.35 | 98.61±0.92 |
| Corn-senesced-green-weeds | | | 99.55±0.42 | 96.26±0.81 |
| Lettuce-romaine-4wk | | | 91.57±2.66 | 88.20±3.58 |
| Lettuce-romaine-5wk | | | 98.90±0.96 | 96.50±1.83 |
| Lettuce-romaine-6wk | | | | 89.65±3.09 |
| Lettuce-romaine-7wk | | | | 75.38±2.75 |
| Vinyard-untrained | | | | 96.49±1.45 |
| Vinyard-vertical-trellis | | | | 99.49±0.59 |
| OA | 98.80±0.54 | 98.16±0.74 (0.64) | 98.25±0.25 (0.09↑) | 96.66±0.34 (1.59) |
| AA | 98.67±0.59 | 97.66±0.62 (1.01) | 97.35±0.47 (0.31) | 94.72±0.49 (2.63) |
| Kappa | 98.32±0.76 | 97.63±0.93 (0.69) | 97.99±0.28 (0.36↑) | 96.28±0.38 (1.71) |

the previous task, and ↑ indicates that the value in that section is higher.

On the IP dataset, the overall accuracy (OA) of the proposed method on tasks 1–4 is 97.56%, 94.07%, 92.57%, and 91.14%,

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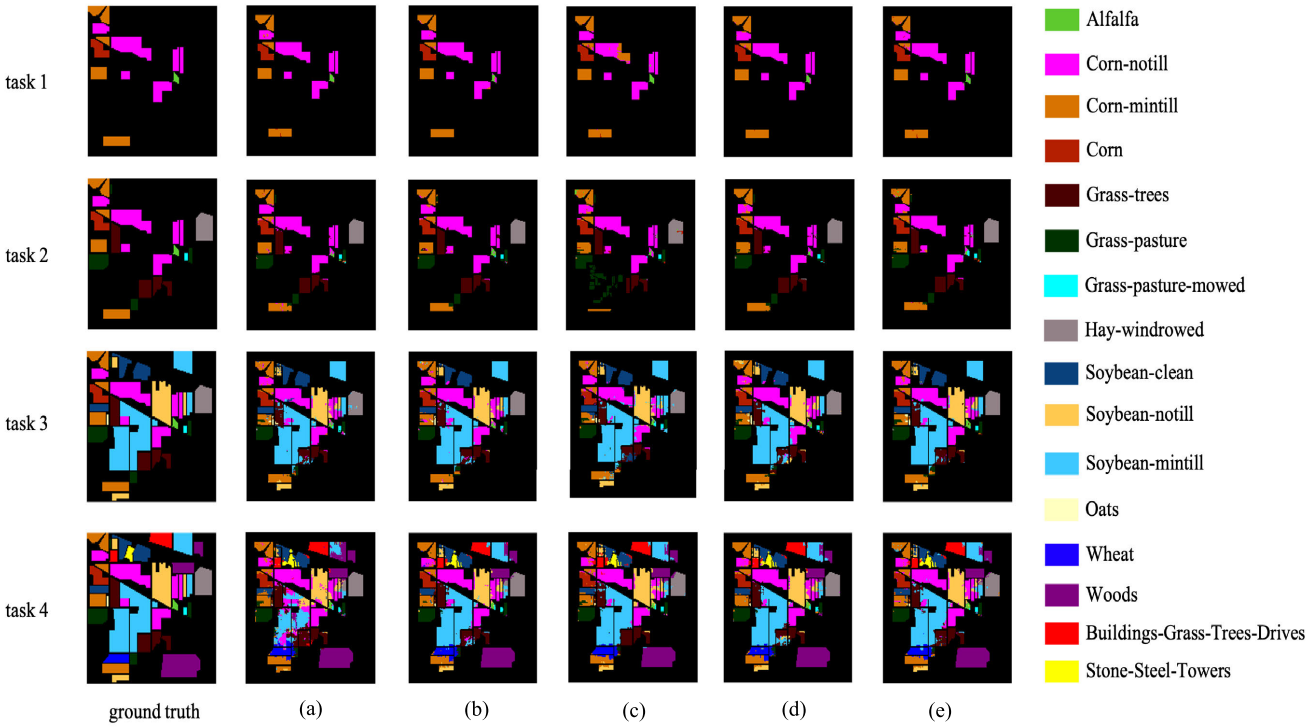



Fig. 6. Class incremental tasks 1–4 classification results map of the comparative experiment on the IP dataset. (a) iCaRL. (b) Bic. (c) WA. (d) CCIL. (e) Ours.

TABLE VII
COMPARATIVE EXPERIMENTAL RESULTS ON THE IP DATASET

| Method | task 1 | task 2 | task 3 | task 4 |
|---|---|---|---|---|
| iCaRL [33] | **96.97** | 91.39 (5.58) | 92.11 (0.72↑) | 81.54 (10.57) |
| Bic [35] | 96.23 | **94.58** (1.65) | **92.87** (1.71) | **90.49** (2.28) |
| WA [37] | 95.17 | **94.15** (1.02) | 89.94 (4.21) | 90.28(0.34↑) |
| CCIL [52] | 96.24 | 92.70 (3.54) | 92.22 (0.48) | 90.45 (1.77) |
| **Ours** | **97.56** | 94.07 (3.49) | **92.57** (1.50) | **91.14** (1.43) |

TABLE VIII
COMPARATIVE EXPERIMENTAL RESULTS ON THE PU DATASET

| Method | task 1 | task 2 | task 3 |
|---|---|---|---|
| iCaRL [33] | **98.64** | 84.35 (14.29) | 85.49 (1.14↑) |
| Bic [35] | 97.24 | **89.73** (7.51) | **89.46** (0.27) |
| WA [37] | 97.28 | 86.90 (10.38) | 87.74 (0.84↑) |
| CCIL [52] | 97.81 | 88.74 (9.07) | 88.11 (0.63) |
| **Ours** | **98.82** | **90.04** (8.78) | **90.06** (0.02↑) |

TABLE IX
COMPARATIVE EXPERIMENTAL RESULTS ON THE SA DATASET

| Method | task 1 | task 2 | task 3 | task 4 |
|---|---|---|---|---|
| iCaRL [33] | 97.86 | 97.39 (0.47) | 96.02 (1.37) | 93.25 (2.77) |
| Bic [35] | **99.07** | 97.25 (1.82) | 97.42 (0.17↑) | 95.81 (1.61) |
| WA [37] | 97.10 | **97.99** (0.89↑) | **97.66** (0.33) | **96.23** (1.43) |
| CCIL [52] | 98.09 | 96.67 (1.42) | 96.12 (0.55) | 95.56 (0.56) |
| **Ours** | **98.80** | **98.16** (0.64) | **98.25** (0.09↑) | **96.66** (1.59) |

respectively. Due to the process of adding new data, the model performs partial forgetting for old knowledge. The OA of the current task compared to the previous task decline is 3.49%, 1.50%, and 1.43%. The OAs on the PU dataset for tasks 1–3 are 98.82%, 90.04%, and 90.06%, respectively. The OAs of the current task compared to the previous task decreases are: 8.78% and 0.02%(↑). On the SA dataset, the OAs on tasks 1–4 are 98.80%, 98.16%, 98.25%, and 96.66%, respectively. The OAs of the current task compared to the previous task decreases are: 0.64%, 0.09%(↑), and 1.59%.

The catastrophic forgetting of previous old class knowledge that affects the global classification effect of the current task is a major concern in the CIL. The experimental results demonstrate that the forgetting level of the model on the IP dataset has a decreasing stable effect on the global classification results. On the PU dataset, the model is affected by catastrophic forgetting to a minor degree. Compared to task 2, the OA of task 3 still increased by 0.02%. On the SA dataset, the overall classification accuracy of the model is relatively stable and less subject to catastrophic forgetting. In particular, compared to task 1, the OA of task 2 still increased by 0.09%.

Extensive experiments have shown that the proposed method ensures continuous learning of new data, without excessive forgetting of the old knowledge already learned. In addition, the proposed method maintains excellent classification results on the final classification task.

### D. Comparative Experiments

To demonstrate the advancedness of the $GS_2FIN$-CIL method, we have selected several CIL methods that are relatively similar to the proposed method framework for extensive comparative experiments. The graphs of the comparative classification results on the three datasets are shown in Figs. 6–8. The OA results of the comparative experiments in class incremental tasks are summarized in Tables VII–IX. We focused on marking the first two OA in each class incremental task.

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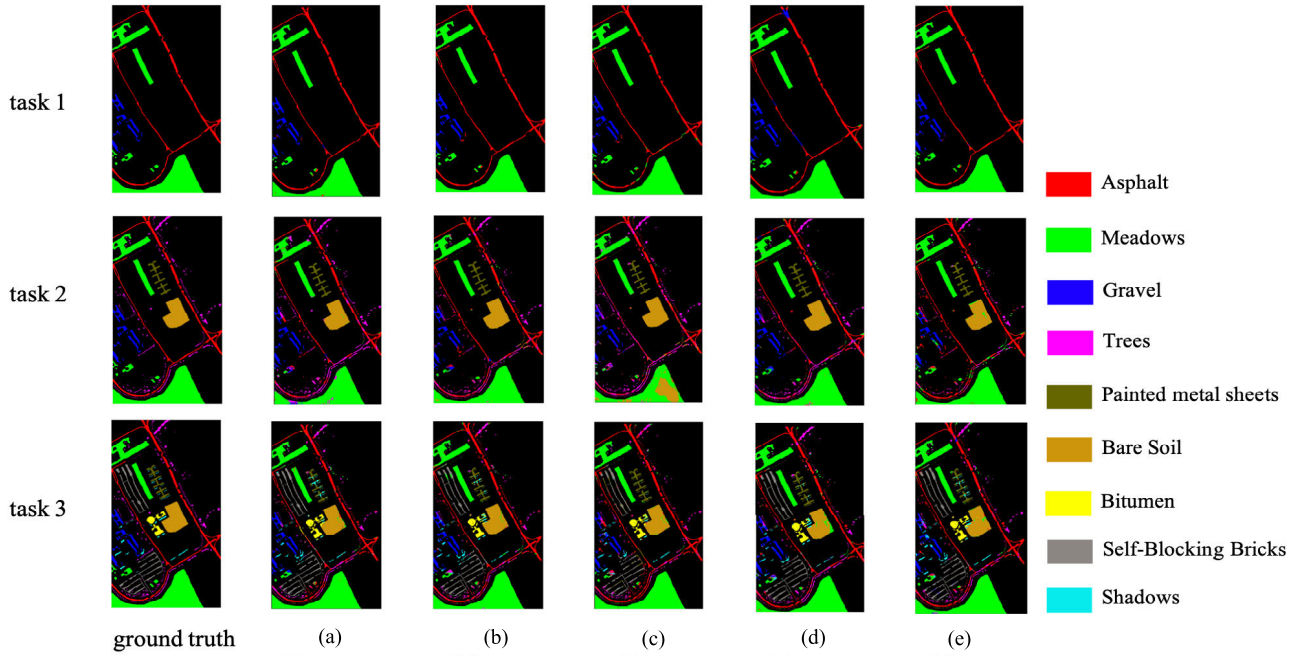



Fig. 7. Class incremental tasks 1–3 classification results map of the comparative experiment on the PU dataset. (a) iCaRL. (b) Bic. (c) WA. (d) CCIL. (e) Ours.

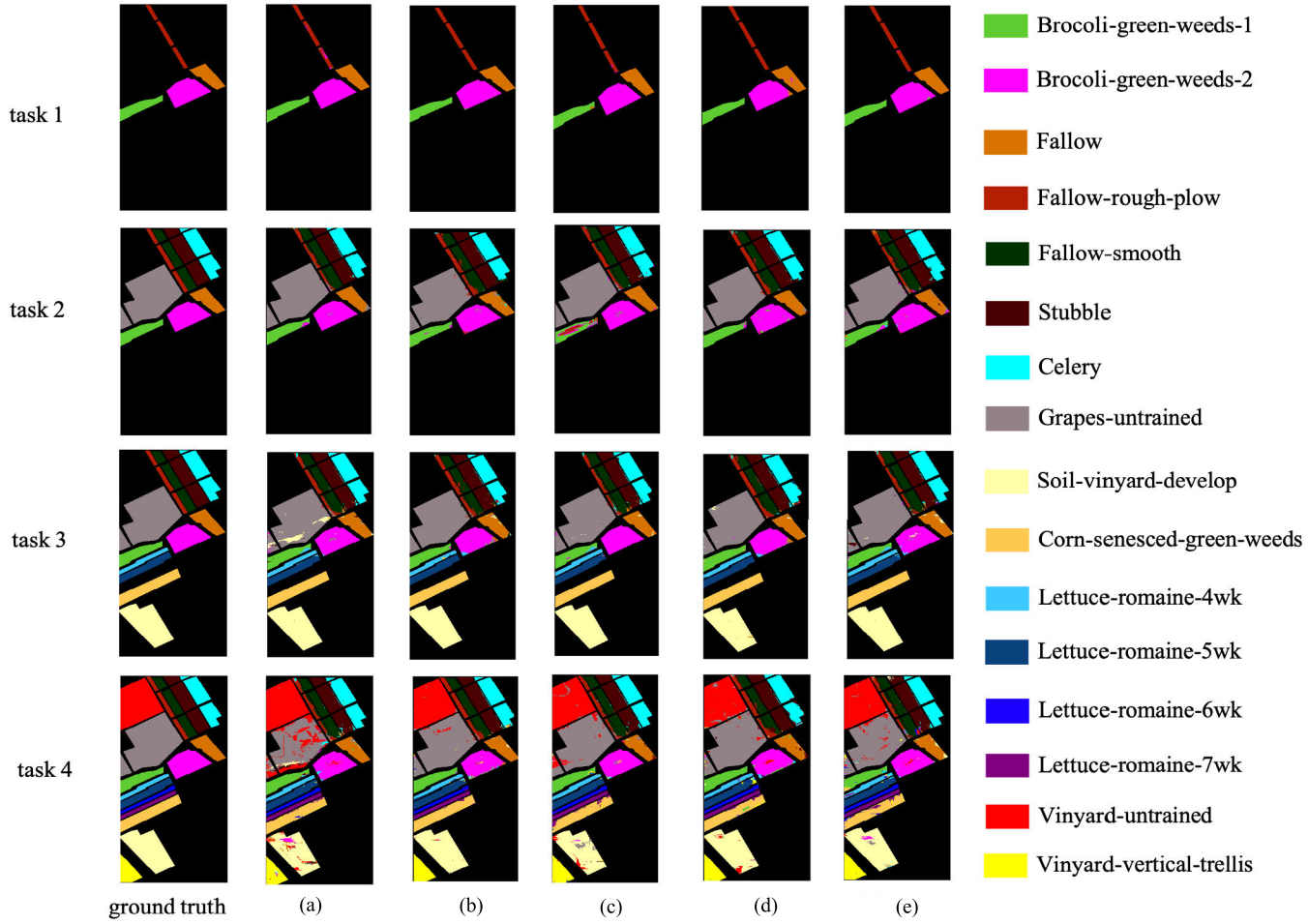



Fig. 8. Class incremental tasks 1–4 classification results map of the comparative experiment on the SA dataset. (a) iCaRL. (b) Bic. (c) WA. (d) CCIL. (e) Ours.

On the IP dataset, the proposed method achieves remarkable OA on tasks 1, 3, and 4, with 97.56%, 92.57%, and 91.14%, respectively. Comparing the OA of the current task with the OA of the previous task, the method has progressively

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TABLE X
COMPARATIVE CLASSIFICATION RESULTS OF TASK 4 ON THE IP DATASET

| Class | Method | | | | |
|---|---|---|---|---|---|
| | iCaRL [33] | Bic [35] | WA [37] | CCIL [52] | Ours |
| Alfalfa | **94.41±6.71** | 94.28±5.06 | 84.94±18.52 | 83.63±20.30 | 91.75±7.32 |
| Corn-notill | 77.88±4.97 | 91.35±0.98 | **91.64±1.26** | 90.39±3.99 | 90.73±1.75 |
| Corn-mintill | 81.48±3.59 | **91.65±2.27** | 89.94±2.29 | 90.11±2.66 | 91.18±2.89 |
| Corn | 93.59±4.23 | 96.27±1.73 | **96.30±1.44** | 95.83±2.43 | 95.25±2.96 |
| Grass-pasture | 83.59±6.18 | 90.51±2.93 | 89.12±4.09 | 90.12±3.53 | **93.07±2.82** |
| Grass-trees | 70.02±3.93 | 86.07±2.74 | 87.53±2.91 | 87.89±4.43 | **89.35±3.29** |
| Grass-pasture-mowed | 90.06±14.98 | **91.79±12.97** | 82.16±15.75 | 77.51±18.76 | 90.67±11.86 |
| Hay-windrowed | 84.20±5.48 | 93.89±1.67 | 94.86±1.26 | 94.45±1.83 | **95.32±1.80** |
| Oats | 90.14±16.56 | 95.50±9.07 | **100.00±0.00** | 76.99±29.30 | 95.00±10.67 |
| Soybean-notill | 85.53±3.98 | **90.65±1.77** | 86.22±2.17 | 88.68±1.66 | 90.28±1.83 |
| Soybean-mintill | **97.59±1.35** | 92.89±1.26 | 94.99±1.69 | 93.22±3.25 | 94.26±1.25 |
| Soybean-clean | 89.19±5.57 | 91.33±1.84 | 91.74±2.77 | **92.46±1.94** | 89.71±3.63 |
| Wheat | 55.38±7.87 | 65.39±3.79 | 64.83±7.71 | **72.19±3.36** | 64.89±4.55 |
| Woods | 78.46±4.89 | 92.44±2.01 | 91.92±2.76 | 94.30±2.00 | **94.56±1.89** |
| Buildings-Grass-Trees-Drives | 85.48±9.08 | 92.32±2.05 | 90.93±3.89 | **93.28±2.15** | 90.57±3.41 |
| Stone-Steel-Towers | 63.03±5.05 | 62.67±3.45 | **63.86±5.67** | 60.50±13.35 | 63.59±4.96 |
| OA | 81.54±2.37 | 90.51±0.62 | 90.28±0.68 | 90.46±1.28 | **91.14±0.59** |
| AA | 82.50±2.79 | 88.69±1.31 | 87.56±1.47 | 86.35±2.64 | **88.72±1.16** |
| Kappa | 79.31±2.59 | 89.19±0.70 | 88.95±0.78 | 89.13±1.46 | **89.91±0.66** |

TABLE XI
COMPARATIVE EXPERIMENTAL RESULTS OF TASK 4 ON THE PU DATASET

| Class | Method | | | | |
|---|---|---|---|---|---|
| | iCaRL [33] | Bic [35] | WA [37] | CCIL [52] | Ours |
| Asphalt | 73.89±9.27 | 78.93±2.70 | 77.47±6.45 | 78.42±5.14 | **80.09±2.20** |
| Meadows | 96.51±1.59 | 97.34±0.61 | 97.91±0.71 | **98.57±0.48** | 97.68±0.39 |
| Gravel | 87.79±2.62 | **89.12±2.39** | 88.89±2.81 | 82.21±12.09 | 86.95±4.68 |
| Trees | 56.11±10.09 | 64.30±6.46 | 65.79±6.36 | 62.87±6.82 | **67.78±4.96** |
| Painted metal sheets | 85.23±2.51 | 83.64±5.84 | 83.36±7.92 | 85.94±0.78 | **86.35±0.41** |
| Bare Soil | 94.51±4.21 | 97.01±1.01 | 91.46±10.86 | 95.28±7.48 | **98.49±0.83** |
| Bitumen | 82.42±6.87 | 85.39±2.79 | 85.08±8.33 | 85.86±2.29 | **88.17±2.65** |
| Self-Blocking Bricks | 88.54±4.19 | **90.35±1.87** | 89.68±3.87 | 86.44±6.21 | 90.04±1.27 |
| Shadows | 45.04±11.16 | **54.55±5.11** | 46.41±8.22 | 48.78±8.56 | 49.61±5.14 |
| OA | 85.49±4.19 | 89.46±0.53 | 87.77±2.20 | 88.11±1.74 | **90.06±0.52** |
| AA | 78.89±2.21 | 82.29±1.28 | 80.67±1.97 | 80.49±1.98 | **82.80±0.87** |
| Kappa | 80.93±5.47 | 86.02±0.70 | 83.91±2.76 | 84.41±2.21 | **86.82±0.68** |

decreasing levels of catastrophic forgetting on tasks 2–4, which are 3.49%, 1.50%, and 1.43%, respectively. In addition, the method achieves the best result of 91.14% on the final classification task 4.

On the PU dataset, compared to the other methods, the proposed method achieves the highest OA on tasks 1–3, with 98.82%, 90.04%, and 90.06%, respectively. On the final classification task 3, our proposed method not only achieves the best classification results but also improves the accuracy by 0.02% compared to the previous task. This demonstrates the ability of the model to alleviate the catastrophic forgetfulness problem while guaranteeing high accuracy.

On the SA dataset, the proposed method achieves remarkable OA on tasks 1–4 and the values tend to be stable. On tasks 2–4, compared to the other methods, the method achieves the highest result of 98.16%, 98.25%, and 96.66%. On task 2, the OA of the method increases by 0.09% compared to the previous task.

The comparative results of the various classes of accuracy on the final classification task are shown in Tables X–XII. The experimental results indicate that the proposed $GS_2$FIN-CIL method achieves the best results of the final classification task on three datasets, which reflects the state-of-the-art progress of the proposed method on CIL.

### *E. Ablation Experiments*

Three modules are mainly designed in the proposed $GS_2$FIN-CIL method, namely, $GS_2$FIAM, BCK classifier, and MLFS. To ensure the validity of these three modules, we performed adequate extensive ablation experiments on the three datasets. The OA of the classification is shown in Tables XIII–XV.

In the ablation experiments, the baseline is the $GS_2$FIN network without the $GS_2$FIAM attention module, the loss function in the training phase is the combination of cross-entropy loss and knowledge distillation loss, the testing phase still uses

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TABLE XII
COMPARATIVE EXPERIMENTAL RESULTS OF TASK 4 ON THE SA DATASET

| Class | Method | | | | |
|---|---|---|---|---|---|
| | iCaRL [33] | Bic [35] | WA [37] | CCIL [52] | Ours |
| Brocoli-green-weeds-1 | 92.29±2.51 | **99.05±0.68** | 94.84±2.94 | 98.51±2.62 | 96.55±1.25 |
| Brocoli-green-weeds-2 | 98.84±1.59 | 99.17±0.44 | 99.32±0.78 | 99.31±0.54 | **99.46±0.46** |
| Fallow | 92.49±6.07 | 93.86±1.86 | 96.07±1.66 | 94.23±2.04 | **96.46±2.11** |
| Fallow-rough-plow | 91.44±4.52 | 84.87±4.91 | **95.37±1.39** | 84.99±7.41 | 95.08±2.36 |
| Fallow-smooth | 94.17±1.39 | 93.17±2.47 | 92.53±3.47 | 93.31±2.21 | **94.53±1.53** |
| Stubble | 94.13±1.39 | 96.05±1.31 | 95.69±1.63 | **97.18±0.70** | 95.78±1.07 |
| Celery | 98.26±1.04 | 97.38±1.13 | **98.27±0.55** | 97.63±0.82 | 98.24±0.64 |
| Grapes-untrained | 99.16±0.49 | 97.31±1.26 | **99.29±0.24** | 98.09±2.42 | 98.89±0.23 |
| Soil-vinyard-develop | 94.66±2.03 | 98.19±1.78 | 98.84±0.97 | **98.98±0.55** | 98.61±0.92 |
| Corn-senesced-green-weeds | 97.24±1.26 | **98.04±1.51** | 95.14±1.43 | 93.44±13.67 | 96.26±0.81 |
| Lettuce-romaine-4wk | 87.32±3.23 | 93.49±3.03 | 88.49±1.51 | **91.18±3.70** | 88.20±3.58 |
| Lettuce-romaine-5wk | 96.39±1.11 | 92.27±2.70 | 95.50±5.16 | 92.32±2.04 | **96.50±1.83** |
| Lettuce-romaine-6wk | 89.03±1.95 | 97.18±1.45 | 89.05±3.86 | **97.80±1.20** | 89.65±3.09 |
| Lettuce-romaine-7wk | 71.93±2.96 | 92.01±2.81 | 76.04±2.85 | **92.69±2.32** | 75.38±2.75 |
| Vinyard-untrained | 83.76±4.06 | 93.66±1.79 | 94.87±1.71 | 92.49±2.32 | **96.49±1.45** |
| Vinyard-vertical-trellis | 96.56±6.79 | 92.73±4.73 | 99.02±1.02 | 96.48±2.57 | **99.49±0.59** |
| OA | 93.25±1.15 | 95.81±0.68 | 96.23±0.53 | 95.56±1.81 | **96.66±0.34** |
| AA | 92.36±1.25 | 94.90±0.59 | 94.27±0.59 | **94.92±1.07** | 94.72±0.49 |
| Kappa | 92.51±1.26 | 95.34±0.76 | 95.80±0.59 | 95.06±2.01 | **96.28±0.38** |

TABLE XIII
ABLATION EXPERIMENTS OF $GS_2$FIN-CIL METHOD ON THE IP DATASET

| | BCK | MLFS | $GS_2$FIAM | task1 | task 2 | task 3 | task 4 |
|---|---|---|---|---|---|---|---|
| Baseline | | | | 98.51±0.35 | 92.34±0.58 | 89.42±0.51 | 85.67±0.78 |
| Baseline+BCK | ✓ | | | **98.65±0.23** | 92.29±0.84 | 89.79±0.56 | 85.95±0.73 |
| Baseline+MLFS | | ✓ | | **98.60±0.14** | 92.39±1.12 | 89.49±0.58 | 85.83±0.62 |
| Baseline+$GS_2$FIAM | | | ✓ | 96.23±3.75 | **94.58±1.24** | **92.87±0.74** | 90.49±0.84 |
| Baseline+BCK+MLFS | ✓ | ✓ | | 98.39±0.49 | 92.87±0.42 | 90.06±0.44 | 86.17±0.67 |
| Baseline+BCK+$GS_2$FIAM | ✓ | | ✓ | 88.46±19.56 | **94.85±1.53** | 92.83±0.43 | **90.67±0.84** |
| Baseline+MLFS+$GS_2$FIAM | | ✓ | ✓ | 95.76±6.59 | 94.57±1.12 | **93.17±0.48** | 90.55±0.53 |
| **Ours** | ✓ | ✓ | ✓ | 97.56±2.26 | 94.07±1.00 | 92.57±1.46 | **91.14±0.58** |

TABLE XIV
ABLATION EXPERIMENTS OF $GS_2$FIN-CIL METHOD ON THE PU DATASET

| | BCK | MLFS | $GS_2$FIAM | task1 | task 2 | task 3 |
|---|---|---|---|---|---|---|
| Baseline | | | | **99.26±0.13** | 91.04±0.37 | 86.34±0.61 |
| Baseline+BCK | ✓ | | | 98.78±1.07 | **91.29±0.38** | 86.69±0.61 |
| Baseline+MLFS | | ✓ | | **99.22±0.23** | **91.07±0.23** | 86.48±0.48 |
| Baseline+$GS_2$FIAM | | | ✓ | 97.24±1.98 | 89.73±1.44 | 89.46±0.53 |
| Baseline+BCK+MLFS | ✓ | ✓ | | 98.92±0.75 | 90.93±0.29 | 87.09±0.42 |
| Baseline+BCK+$GS_2$FIAM | ✓ | | ✓ | 97.64±2.64 | 90.04±1.42 | 89.92±0.60 |
| Baseline+MLFS+$GS_2$FIAM | | ✓ | ✓ | 98.19±1.54 | 90.14±1.07 | **89.97±0.54** |
| **Ours** | ✓ | ✓ | ✓ | 98.82±0.51 | 90.04±1.96 | **90.06±0.52** |

TABLE XV
ABLATION EXPERIMENTS OF $GS_2$FIN-CIL METHOD ON THE SA DATASET

| | BCK | MLFS | $GS_2$FIAM | task1 | task 2 | task 3 | task 4 |
|---|---|---|---|---|---|---|---|
| Baseline | | | | 99.50±0.67 | 97.27±0.25 | 96.77±0.41 | 93.57±0.57 |
| Baseline+BCK | ✓ | | | **99.62±0.29** | 97.49±0.25 | 97.11±0.32 | 94.39±0.41 |
| Baseline+MLFS | | ✓ | | 99.49±0.71 | 97.38±0.25 | 96.85±0.23 | 93.86±0.43 |
| Baseline+$GS_2$FIAM | | | ✓ | 99.07±0.94 | 97.25±0.45 | 97.42±0.24 | 95.81±0.81 |
| Baseline+BCK+MLFS | ✓ | ✓ | | **99.55±0.18** | **98.52±0.13** | **98.28±0.12** | 95.48±0.36 |
| Baseline+BCK+$GS_2$FIAM | ✓ | | ✓ | 98.64±1.11 | 97.02±1.24 | 97.56±0.21 | 95.95±0.94 |
| Baseline+MLFS+$GS_2$FIAM | | ✓ | ✓ | 96.80±3.03 | 97.08±0.34 | 97.61±0.24 | **96.22±0.35** |
| **Ours** | ✓ | ✓ | ✓ | 98.80±0.54 | **98.16±0.73** | **98.25±0.24** | **96.66±0.34** |

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only cross-entropy loss, and the designed BCK classifier is not utilized.

On the class incremental task 4 of the IP dataset, the OA increases by 0.28% compared to baseline after using the BCK classifier alone, 0.16% compared to baseline after using MLFS in the training phase, and 4.82% compared to baseline after adding the $GS_2$FIAM module to the network. Finally, the proposed method improves the OA compared to baseline by 5.47%. On the class increment task 3 of the PU dataset, the OA increases by 0.35% compared to baseline after using the BCK classifier alone, 0.14% compared to baseline after using MLFS in the training phase, and 3.12% compared to baseline after adding the $GS_2$FIAM module to the network. Finally, the proposed method improves the OA compared to baseline by 3.72%. On the class increment task 4 of the SA dataset, the OA increases by 0.82% compared to baseline after using the BCK classifier alone, 0.29% compared to baseline after using MLFS in the training phase, and 2.24% compared to baseline after adding the $GS_2$FIAM module to the network. Finally, the method improves the OA compared to baseline by 3.09%.

The experimental results adequately illustrate that the three proposed modules have a significant contribution to the classification result enhancement. Among them, the $GS_2$FIAM attention mechanism module has the most remarkable improvement in the network.

## V. Conclusion

In this article, we strive to develop a new CIL method called $GS_2$FIN-CIL for HSIC. The proposed method is able to continuously learn additional class data and alleviates the notorious catastrophic forgetting problem in CIL. We employ the attention mechanism in the CIL process and design the $GS_2$FIAM module in the network, which considers the 3-D data characteristics of HSIs and performs intensive integration extraction of spatial–spectral features from the geometric space perspective. The BCK classifier is designed to balance the knowledge of old and new classes learned by the model, thus alleviating the problem of proximate causality bias for new classes. We propose a novel MLFS in the proposed method, which is able to memorize past knowledge during the training process of continuous iteration of the network, thus reducing the forgetting of old classes' knowledge by the model. In addition, the proposed method is capable of achieving superior classification results with the addition of minor new class data within a limited sample space.

Specifically, CIL in the field of HSIC leaves plenty of space for exploration. In the future, we would endeavor to employ few-shot learning to maintain the memory of learned knowledge in CIL while using only few samples, thus occupying less sample space and achieving lifelong learning without forgetting.

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